

Solution

National University
of Computer & Emerging Sciences, Peshawar Campus



Program/Section: BAI-2A	Examination: Quiz # 02
Semester: Spring-2026	Total Marks: 10, Weightage: 2.5
Course: MT1008 (Multivariable Calculus)	Date:
Name:	Reg. No:

Time allowed: 20 minutes

1. Let $f(x, y) = x^2y + y^3$. Find a unit vector at the point $(1, 2)$ for which the directional derivative of f is maximum and compute that maximum value.
2. Find partial derivatives using definition for the following function:

$$f(x) = \begin{cases} \frac{x^2y}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Q1. $f_x = 2xy$, $f_y = x^2 + 3y^2$
 $\Rightarrow \nabla f = \langle f_x, f_y \rangle = \langle 2xy, x^2 + 3y^2 \rangle$
 $\Rightarrow \nabla f(1, 2) = \langle 2(1)(2), 1^2 + 3(2)^2 \rangle$
 $= \langle 4, 13 \rangle$

Directional derivative of f is maximum if $\vec{u}$ and $\nabla \vec{f}$ are parallel.

So, $\vec{u} = \frac{\nabla \vec{f}}{|\nabla \vec{f}|} = \frac{\langle 4, 13 \rangle}{\sqrt{4^2 + 13^2}} = \left\langle \frac{4}{\sqrt{185}}, \frac{13}{\sqrt{185}} \right\rangle$

and $D_{\max} f = |\nabla \vec{f}| = \sqrt{4^2 + 13^2} = \sqrt{185}$

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Q2. By def. $f_x = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$

$$\Rightarrow f_x \stackrel{\lim_{h \rightarrow 0}}{=} \frac{1}{h} \left[\frac{(x+h)^2 y}{(x+h)^2 + y^2} - \frac{x^2 y}{x^2 + y^2} \right]$$

$$\stackrel{\lim_{h \rightarrow 0}}{=} \frac{1}{h} \left[\frac{y(x+h)^2 \cdot (x^2 + y^2) - x^2 y [(x+h)^2 + y^2]}{[(x+h)^2 + y^2] \cdot (x^2 + y^2)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{y}{h} \left[\frac{\cancel{x^2(x+h)^2} + y^2(x+h)^2 - \cancel{x^2(x+h)^2} - x^2 y^2}{[(x+h)^2 + y^2] \cdot (x^2 + y^2)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{y}{h} \left[\frac{\cancel{x^2 y} + h^2 y^2 + 2xh y^2 - \cancel{x^2 y^2}}{[(x+h)^2 + y^2] \cdot (x^2 + y^2)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{y}{h} \left[\frac{h(h y^2 + 2x y^2)}{[(x+h)^2 + y^2] \cdot (x^2 + y^2)} \right] = \frac{2x y^3}{(x^2 + y^2)^2}$$

Also,

$$f_y = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{x^2(y+h)}{x^2 + (y+h)^2} - \frac{x^2 y}{x^2 + y^2} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{\cancel{x^4 y} + x^4 h + x^2 y^2 h + \cancel{x^2 y^3} - \cancel{x^4 y} - \cancel{x^2 y^3} - x^2 h^2 y - 2x^2 y^2 h}{[x^2 + (y+h)^2] \cdot (x^2 + y^2)} \right]$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{x^4 - x^2 y^2 - x^2 h y}{[x^2 + (y+h)^2] \cdot (x^2 + y^2)} \right] = \frac{x^4 - x^2 y^2 - x^2(0)y}{[x^2 + (y+0)^2] \cdot (x^2 + y^2)}$$

$$= \frac{x^2(x^2 - y^2)}{(x^2 + y^2)^2}$$

